

Matrix functions and stability

A dimension-independent polynomial norm bound from super-identical pseudospectra

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Topic: Polynomial matrix norms and pseudospectral information

Rating rationale: Existing information controls polynomial norms with a dimension-dependent constant; a uniform comparison would strengthen pseudospectral inference about nonnormal dynamics and polynomial matrix functions.

Resolution: negative, 15 September 2026

Solved negatively by Georg Maierhofer (University of Cambridge). The manuscript is dated 14 September 2026 and passed a fresh independent informal AI-agent audit on 15 September 2026. The exact dimension-dependent constants and their optimal growth remain open.

The [counterexample manuscript](#), **Theorem 1 and equation (15)**, constructs real nonnegative nilpotent weighted shifts of order $N = (m + 1)^2$, for every integer $m \geq 2$ and real $t > 1$. All singular values agree after every complex scalar shift. For the common polynomial $p_m(z) = \sum_{j=1}^m z^{(m+2)j}$, it proves

$$\frac{\|p_m(X_{m,t})\|_2}{\|p_m(Y_{m,t})\|_2} \geq \frac{\sqrt{m}}{1 + (m-1)/t^2}, \quad p_m(Y_{m,t}) \neq 0.$$

Taking $t = m$ gives a ratio at least $(4/5)\sqrt{m}$, so the original uniform-boundedness target is refuted.

Corollary 3 gives $C_N \geq \sqrt{\lfloor \sqrt{N} \rfloor - 1}$ for $N \geq 9$. It does not determine the sharp C_N or close the gap to the known upper bound.

Evidence level: a complete written argument with reproducible exact and numerical checks, developed and self-reviewed with AI assistance. The [fresh maintainer audit](#) confirms the complete negative resolution and records independent mathematical review and fresh checks. The [submitted informal Codex AI-agent audit](#) is also retained. Those original audits were informal; the separate Lean verification is documented below. No external human peer review is asserted. See the [source, checks and provenance](#). The existing difficulty and importance ratings are retained as historical ratings of the original boundedness question.

Lean proof and verification evidence

Formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. Georg Maierhofer retains credit for the mathematical counterexample above. The [complete Lean proof at immutable revision 208e30d8](#) pins Lean 4.33.1, Mathlib 0df444a360eaa60ab8c11dca51a86af692955474 and LeanCert 621a43d7cf21f87872392a01e874f2f1dbddc926.

The declarations in namespace `NLA.MF24` include:

- `family_super_identical`: equality of every ordered singular value after every complex shift.
- `polynomial_norm_ratio` and `finite_rational_family_ratio`: the explicit norm-ratio estimates for the original weighted-shift family.
- `arbitrarily_large_ratios`: a finite-dimensional pair and polynomial exceeding every proposed real comparison bound.
- `no_uniform_comparison`: the complete negative answer to the original uniform-boundedness question.

All 22 declarations, including their full determinant, spectrum and operator-norm dependencies, are listed in [formalization.yaml](#) and checked against the independently reviewed [Challenge](#). The definitions use genuine complex polynomial evaluation and the induced norm on complex Euclidean space. Singular-value equality includes multiplicities and zeros; no normality, fixed dimension or restricted shift is assumed. The formal estimate uses the conservative denominator $t^2 + m$, giving $(2/3)\sqrt{m}$ at $t = m$. This still refutes the complete original target. The manuscript’s sharper constant, dimension-padding corollary and optimal growth question are not claimed as Lean-verified results.

On 15 September 2026, the [actual GitHub Linux verification job](#) checked this exact proof commit. All 22 exports passed LeanCert kernel assertions, real non-root Comparator statement matching, default-kernel replay and transitive axiom checks. Only `propext`, `Classical.choice` and `Quot.sound` support the target results. Invalid-proof, mismatched-statement, forbidden-axiom and sandbox controls passed. The [retained raw logs, artifact and input hashes](#) bind that execution to all 192 candidate project files. These are actual GitHub Linux checks; no local macOS Lean execution is claimed.

Two independent AI-agent referees accepted the complete mathematical source and reviewed the actual runtime evidence: [referee 1](#) and [referee 2](#). They applied the repository’s scoped Tau Ceti protocol. This is separate from external human peer review or certification that the checker software is infallible.

To reproduce the ordinary proof build, enter [the Lean project directory](#) with Lean installed and run:

```
lake exe cache get && lake build
```

The default target is the complete Solution. For the additional sandboxed Comparator, kernel replay and rejection checks, follow the [shared Linux verification instructions](#). Solution never imports the Challenge’s deliberate specification placeholders. This evidence names the exact tested proof revision; subsequent documentation commits have separate published-commit checks.

Context and notation

All matrices are complex and all norms are spectral norms. Write $s_1(M) \geq \dots \geq s_N(M)$ for the singular values of M . Define

$$A \sim_{\text{sip}} B \iff s_j(A - zI) = s_j(B - zI) \quad (z \in \mathbb{C}, 1 \leq j \leq N).$$

Define the sharp comparison constant

$$C_N = \sup \left\{ \frac{\|p(A)\|_2}{\|p(B)\|_2} : A, B \in \mathbb{C}^{N \times N}, A \sim_{\text{sip}} B, p \in \mathbb{C}[z], p(B) \neq 0 \right\}.$$

As recorded in [SP-15](#), super-identical pseudospectral data also imply ordinary similarity, so $p(A) = 0$ if and only if $p(B) = 0$.

Problem statement

Determine whether $\sup_{N \geq 1} C_N < \infty$. Equivalently, does there exist an absolute constant C such that

$$\|p(A)\|_2 \leq C \|p(B)\|_2$$

for every dimension, every super-identical-pseudospectral pair and every polynomial? Determining the sharp C_N is the closely related quantitative question, retained here rather than split into another entry.

References

- Fortier Bourque and Ransford, *Super-identical pseudospectra*, p. 513, immediately after Theorem 1.3, explicitly ask both whether their $\sqrt{N}$ bound is optimal and whether a dimension-independent bound exists. The sharp-constant notation above is editorial; the boundedness question is source-stated.
- Thomas Ransford and Nathan Walsh, *A four-mean theorem and its application to pseudospectra*, [arXiv:2109.14472v2](#), 9 July 2022, Theorem 1.3, prove

$$\|p(A)\|_2 < \sqrt{N-2} \|p(B)\|_2 \quad (N \geq 4)$$

unless both sides vanish. **Proposition 5.1** and its following argument show sharpness for $N = 4$: $C_4 = \sqrt{2}$. **Theorem 1.4** concerns unbounded condition numbers of similarity transforms in fixed dimension, a different quantity.

Thus $C_N \leq \sqrt{N-2}$ for $N \geq 4$. The small-dimensional unitary/transpose classification gives $C_1 = C_2 = C_3 = 1$.

Original scope and literature check

The following is the repository's pre-claim literature record, retained unchanged.

The unknown absolute constant, not the superseded $\sqrt{N}$ sharpness guess, is the admission target. This comparison does not follow just from ordinary similarity, since a badly conditioned similarity can change norms substantially. Searches through 2026-09-11 for super-identical polynomial norm bounds, dimension-independent bounds, sharp constants and successors to the four-mean paper found no full solution. The current arXiv version history still ends at v2, 9 July 2022. This is an older-source question supported by bounded searches; no 2026 explicit open-status reaffirmation was located.